

Language Modeling

- A language model is a probability distribution over sequences of words. Given any sequence of words of length m , a language model assigns a probability $P(w_1, \dots, w_m)$ to the whole sequence. Language modeling (LM) is the use of various statistical and probabilistic techniques to determine the probability of a given sequence of words occurring in a sentence.
- Language models generate probabilities by training on text corpora in one or many languages.
- Given that languages can be used to express an infinite variety of valid sentences, language modeling faces the problem of assigning non-zero probabilities to linguistically valid sequences that may never be encountered in the training data.
- Several modeling approaches have been designed to surmount this problem, such as recurrent neural networks or transformers.
 - A recurrent neural network (RNN) is a class of artificial neural networks where connections between nodes can create a cycle, allowing output from some nodes to affect subsequent input to the same nodes.
 - A transformer is a deep learning model that adopts the mechanism of self-attention, differentially weighting the significance of each part of the input data. Like recurrent neural networks (RNNs), transformers are designed to process sequential input data, such as natural language. Unlike RNNs, transformers process the entire input all at once. The attention mechanism provides context for any position in the input sequence. (Does not process the input sentence one word at a time).

- Neural language models (or *continuous space language models*) - that make use of neural networks - use continuous representations or embeddings of words to make their predictions.
- Neural networks represent words in a distributed way, as non-linear combinations of weights in a neural net.
- Typically, neural net language models are constructed and trained as probabilistic classifiers that learn to predict a probability distribution over the vocabulary (V), given some linguistic context:

$$P(w_t | context) \forall t \in V$$

- A notable language model is BERT: Bidirectional Encoder Representations from Transformers.
- BERT is a transformer-based machine learning technique for natural language processing (NLP) pre-training developed by Google.
- Note that bidirectional representations condition on both pre- and post- context (e.g., words) in all layers.

- **APPROACH - First Pre-train then Fine-tune:**
 - Here, the goal of pre-training is to leverage large amounts of unlabeled text and build a general model of language understanding before being fine-tuned on various specific NLP tasks such as machine translation, text summarization etc.
- **Popular pre-training scheme: Masked Language Modeling (MLM)**
 - Under Masked Language Modelling, we typically mask a certain % of words in a given sentence and the model is expected to predict those masked words based on other words in that sentence. Such a training scheme makes this model bidirectional in nature because the representation of the masked word is learnt based on the words that occur it's left as well as right.
- **BERT: is a pre-trained language model, which proposed a new pre-training objective that enables the training of a deep bi-directional transformer network. This objective is called Masked Language Modeling (MLM).**
- **BERT: extremely effective on many NLP tasks, including sentiment analysis and emotion detection.**
- **However: the proposed pre-training objectives of BERT do not induce any sentiment or emotion-specific biases into the model.**
- **In general: the pre-training does not take into account the down-stream task on which the model will be applied.**
- **Today's task: inducing information from emotion or sentiment lexicons into the BERT pre-training objective to improve the performance on tasks from sentiment analysis and emotion detection.**

eMLM: A New Pre-training Objective for Emotion Related Tasks

(Sosea and Caragea, 2021)

- **Sosea and Caragea (2021) present Emotion Masked Language Modeling (eMLM), a variation of Masked Language Modeling, aimed at improving the BERT language representation model for emotion detection and sentiment analysis tasks.**
- **Pre-training corpora: the same as for the original BERT model (Wikipedia and BookCorpus)**
 - ✓ **BookCorpus is a large collection of free novel books written by unpublished authors, which contains 11,038 books (around 74M sentences and 1G words) of 16 different sub-genres (e.g., Romance, Historical, Adventure, etc.).**

Approaches:

- ✓ **Emotion lexicons have been used to produce features which can be used inside a machine learning algorithm to improve the performance on emotion detection tasks (Mohamma, 2012; Sykora et al., 2013; Biyani et al., 2014; Khanpour and Caragea, 2018).**
- ✓ **Now: instead of leveraging these lexicons to design features, the lexicons are used to obtain language representations that are more suitable for emotion and sentiment tasks.**

- **The technique:** different masking probabilities - guided by sentiment and emotion lexicons - for the BERT pre-training procedure are studied.
 - eMLM does not uniformly mask the tokens in the input sequence. The model leverages lexicon information and assigns higher masking probabilities to words that are more likely to be important in sentiment or emotion contexts.
- **Training:** the eMLM model is trained in the same fashion as the vanilla BERT (pre-training on Wikipedia and BookCorpus).
- **Testing:**
 - **DATASETS:** Stanford Sentiment Treebank, GoEmotions, CancerEmo
 - **LEXICONS:** EmoLex

- **OUR GOAL: task-guided unsupervised pre-training**
- **The objectives of the BERT pre-training:**
 1. **Masked Language Modeling (MLM) randomly masks 15% of tokens in a sequence, followed by a supervised prediction of the masked tokens.**
 2. **Next Sentence Prediction (NSP) predicts in a binary fashion if two sentences follow each other.**

- To be used in various downstream tasks, BERT has to undergo a task-specific fine-tuning step – where the contextualized embedding is adapted to the needed task.
- HERE: the downstream performance is improved by focusing on the target task in the pre-training phase as well.
- The conclusion: *task-guided unsupervised pre-training* helps the performance considerably (concerning emotion related tasks).

Masking Emotion Words

➤ We introduce:

Emotion Masked Language Modeling (eMLM) $\equiv$ a variation of MLM targeted at inducing emotion or sentiment-specific biases in the BERT pre-training phase.

➤ Technique:

- BERT uses a uniform probability (15%) to mask the tokens in an input sentence;
- HERE: assign higher probabilities to tokens which are emotionally rich words from an available lexicon L . Denote this probability by k , which is a hyperparameter in the eMLM method.

The masking process:

Input: sentence S

- 1. Extract the words that belong to the lexicon L ; denote them by E ;**
- 2. Set the masking probability of these words as $P(w_e)=k$, $\forall w_e \in E$;**
- 3. To ensure we mask 15% of the words in total, lower the masking probability of the non-emotionally-rich words using the formula**

$$P(w_n) = \frac{\max(|S| \cdot 0.15 - |E| \cdot k, 0)}{|S| - |E|}, \forall w_n \notin E$$

where $|\cdot|$ represents the size of a set.

➤ **Note that:**

- The rest of the **training process** is the same as the original BERT pre-training (i.e. train the BERT model from scratch using eMLM and NSP on the same datasets, Wikipedia and BookCorpus).
- **Whole word masking** is used both for eMLM and the MLM (i.e. all subtokens corresponding to a word are masked).

➤ **Examples** of how new masked probabilities change from MLM to eMLM are shown in Table 1 (next slide).

SENT	They	look	absolutely	perfect	together	I	hope	its	that	way	in	real	life	too
MLM	0.15	0.15	0.15	0.15	0.15	0.15	0.15	0.15	0.15	0.15	0.15	0.15	0.15	0.15
eMLM	0.09	0.09	0.09	0.50	0.09	0.09	0.50	0.09	0.09	0.09	0.09	0.09	0.09	0.09
SENT	Most	tiring	thing	was	the	drive	one	hour	each	way				
MLM	0.15	0.15	0.15	0.15	0.15	0.15	0.15	0.15	0.15	0.15				
eMLM	0.11	0.50	0.11	0.11	0.11	0.11	0.11	0.11	0.11	0.11				

Table 1. Comparison of masked probabilities between MLM and eMLM on two example sentences

Example (from Table 1)

In the first example, there are two emotion words, *perfect* and *hope*, for which a masking probability of $k = 0.50$ was used. So:

- **The probabilities of these two words are set to 50%.**
- **The non emotionally-rich word probability is lowered from 15% to 9%, in order to keep the sum of probabilities constant.**

EXPERIMENTS

- **Experimental setup**
- **Datasets**
- **Lexicons**

Experimental Setup

- **Various benchmark datasets from sentiment analysis and emotion detection are used (see next slide).**
- **For every dataset considered the provided training, validation, and test splits are used.**
- **For statistical significance, each model is fine-tuned 10 times with different random seeds and the average F1 score is reported.**
- **Various masking probabilities, k , are investigated. In the experiments k ranges from 0.2 to 1.0. The finding is that 0.5 works best.**

Datasets

Benchmark datasets:

1. Stanford Sentiment Treebank (SST) (Socher et al.,2013) contains 11,855 sentences from movie reviews, annotated with 5 sentiment labels: negative, somewhat negative, neutral, somewhat positive and positive.

Note that: first, the tests consider the binarized dataset, called SST-2, where the examples with the *negative* and *somewhat negative* labels are merged into a *negative* class, and the examples with the *somewhat positive* and *positive* labels are merged into a positive class (with *neutral* class being removed). Second, the experiments consider the SST fine-grained version (SST-5), which uses all 5 labels.

2. **GoEmotions** (Demszky et al., 2020): a sentence-level multilabel dataset of 58,000 comments curated from Reddit and annotated with 27 emotion categories and the neutral class.
3. **CancerEmo** (Sosea and Caragea, 2020): a sentence-level multilabel dataset of 8,500 sentences labeled with the 8 Plutchik (Plutchik, 1980) basic emotions from an Online Health Community for ill people.

➤ **Note that, in the performed experiments, the emotion or sentiment granularity is varied from 2 classes to as many as 28 classes.**

Lexicons

- **EmoLex (Mohammad and Turney, 2013) - a lexicon of :**
 - ✓ **6,000 words associated with 8 Plutchik basic emotions (*sadness, anger, joy, surprise, anticipation, trust, fear, disgust*) and**
 - ✓ **5,555 words associated with the *positive* and *negative* sentiments.**
- **Note that the sentiment and emotion words are considered separately in order to analyze the impact of each on the performance of eMLM.**
- **Notations:**
 - **eMLM (E) $\equiv$ the approach which masks the emotion-revealing words;**
 - **eMLM (S) $\equiv$ the approach which masks the sentiment-revealing words.**

Results

See the results of the experiments in:

Sosea Tiberiu and Caragea Cornelia, eMLM: A New Pre-training Objective for Emotion Related Tasks. Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Short Papers), pp. 286-293. August 1-6, 2021 (online).

Conclusions

- **A new BERT pre-training objective suited for sentiment analysis and emotion detection tasks was introduced.**
- **This approach needs no additional pre-training compared to the vanilla BERT.**
- **This approach improves the performance by 1.2% F1 on average on various tasks.**

Suggested future research:

- **Studying different lexicons outside the sentiment analysis and emotion detection domains to investigate if the model generalizes well on other domains (e.g. financial)**
- **Studying lexicons from a related domain: hate speech detection, for which see**

Dascalu Stefan, Hristea Florentina, Towards a Benchmarking System for Comparing Automatic Hate Speech Detection with an Intelligent Baseline Proposal. Mathematics, 2022, 10(6), 945;

<https://doi.org/10.3390/math10060945>